

Dissertation Proposal Form

PDE4445

Please fill out this form with the details for your project ideas. Refer to the appendix for clarification regarding the various terms.

Once finalized, you will also have to consider the ethical guidelines for the project. Typically, if your evaluation strategy involves human participants, you will be required to get prior ethical approval. Based on your understanding of the project and discussions with supervisor, please select the declaration that applies to you regarding ethical approval.

Student Details:

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Project Details:

Proposed Title	Trolley-X: An Autonomous, Retrofittable Follow-Me Cart for Multi-Payload Transport
Keywords	Smart cart, ROS 2, autonomous mobility, follow-me, UWB positioning, Kalman filter, LiDAR safety, sensor fusion, path planning, retrofittable payload platform
Problem Definition (Maximum 500 words)	Across many environments - warehouses and retail back-of-house, hospitals, agricultural plots, events and workshops - people still move loads by manually pushing or pulling carts. This is physically demanding, inefficient and ties up labour; it is especially difficult for elderly or mobility-limited operators and over long or repeated trips. The core engineering problem is the absence of an affordable, open, autonomous "follow-me" cart that can be retro-fitted to carry a variety of payloads while safely navigating around people and obstacles in dynamic, unstructured indoor environments. Commercial follow-carts exist (e.g. Foxtech FOLO industrial follow-carts) but are closed, fixed-configuration platforms; retail smart carts (Amazon Dash Cart, Caper) focus on self-checkout, still require the user to push them, and are not built on an open, ROS 2-based architecture suitable for adaptation and integration. Project Trolley-X addresses this by developing an autonomous, retrofittable smart cart that follows its operator hands-free using ultra-wideband (UWB) positioning fused with a Kalman filter, and enforces a three-zone LiDAR-based safety system (warn / slow / stop) for collision-free operation around people and obstacles. A modular payload platform replaces the fixed shopping basket so the same chassis can carry different loads. The project builds on a fully validated ROS 2 Jazzy simulation (submitted as PDE4445 Coursework 2) and converts it into a physical, self-funded prototype for lab validation. Vision-based item recognition and self-checkout are explicitly out of scope for this prototype; the focus is robust autonomous mobility and safety. Time permitting, an autonomous return-to-home path-planning capability will be added as an enhancement.
Research Question(s)	The questions we are trying to answer: • How can UWB positioning fused with a Kalman filter achieve robust, smooth hands-free operator-following at a maintained 1.0 m distance in

	dynamic indoor environments? • To what extent can a three-zone LiDAR-based safety system (warn / slow / stop) detect people and obstacles and prevent collisions in crowded, unstructured spaces? • What are the key performance differences (accuracy, latency, reliability, safety) between the validated ROS 2 simulation and the physical hardware prototype? • (Enhancement / if time permits) To what extent can the platform perform autonomous return-to-home path planning after a payload is delivered?
Background Review (Maximum 500 words)	Autonomous mobile robots (AMRs) for material transport are well established in warehouse logistics, and “follow-me” carts have emerged for both industrial and retail use. Commercial examples include Foxtech’s FOLO industrial follow-carts (UWB/LiDAR-based following for heavy loads) and retail smart carts such as Amazon Dash Cart and Caper (acquired by Instacart for USD 350 million). The retail systems focus on computer-vision self-checkout but require the shopper to push the cart, while industrial follow-carts are typically closed, fixed-configuration products. Few combine open architecture, retrofittable multi-payload carrying and autonomous following. Technically, indoor robot navigation commonly fuses LiDAR, UWB and odometry using filtering techniques such as the Kalman filter for localisation, with ROS / ROS 2 as the de-facto middleware for modular robotic architectures. LiDAR is widely used for obstacle detection and zoned (warn/slow/stop) safety, while UWB provides robust range/position to a carried tag for follow-me behaviour and is far less sensitive to lighting than vision. Most prior work targets warehouses or controlled environments rather than crowded, unstructured spaces with random pedestrian movement. The team has already developed and validated a complete ROS 2 Jazzy simulation in Gazebo Harmonic (four-package architecture: behaviour, navigation, description, gazebo) featuring UWB + Kalman following, three-zone LiDAR safety and a world with dynamic pedestrians. This work (public GitHub repository and demonstration video) confirms feasibility in simulation. The remaining gaps are physical hardware integration, real-world sensor-noise handling, robust UWB ranging hardware and a cost-effective, self-funded BOM. Trolley-X positions itself as an open, ROS 2-native, autonomous, retrofittable follow-me cart combining mobility and safety - distinct from closed industrial follow-carts and from push-based retail smart carts. Vision-based item recognition is deliberately excluded from this prototype to keep the build affordable and the engineering focused on autonomous mobility and safety.
Aims	To design, implement, integrate and evaluate a physical prototype of an autonomous, retrofittable follow-me cart (Trolley-X) that provides hands-free operator-following and collision-free navigation around people and obstacles, thereby demonstrating a complete robotic system integration solution. Vision-based item recognition / self-checkout is out of scope; autonomous return-to-home is an optional enhancement.
Objectives (Bullet points)	• Review current autonomous and follow-me cart technologies, sensor-fusion techniques (UWB, LiDAR, odometry) and zoned-safety methods. • Extend the existing validated ROS 2 Jazzy simulation with hardware-in-the-loop considerations. • Procure components and fabricate/assemble the physical aluminium-chassis prototype with a modular payload platform, per the revised self-funded BOM. • Port and integrate the UWB-Kalman follower and three-zone LiDAR safety (person/obstacle detection and braking) from simulation to physical

	hardware. • Implement an autonomous return-to-home path-planning capability (enhancement, if time permits). • Perform quantitative evaluation of following accuracy, safety performance and system reliability in controlled lab conditions. • Document the complete system, produce a demonstration video, and prepare the dissertation and open-source release.
Scope of Work (per team member)	Each member owns a distinct research question, justifies their design choices against alternatives using literature and their own experiments, tests them rigorously, and communicates directly with the supervisor on their own strand. Ashwin Murali Thanalapati (M01037932) — Sensor fusion & control. Research question: How does fusing UWB ranging with a Kalman filter improve the robustness and smoothness of hands-free following compared with alternative positioning methods, and how reliably does a three-zone LiDAR layer prevent collisions? • Why UWB + Kalman: UWB time-of-flight gives centimetre-level ranging that is robust to lighting and less prone to the drift and multipath of RSSI methods, while the Kalman filter smooths noisy range and estimates velocity for stable following — to be justified against BLE/RSSI, vision-based tracking and VHF beaconing [cite: ...]. • Validation: repeated controlled trials ($\geq N$ runs) measuring distance-holding error against the 1.0 m setpoint, latency and jitter; a baseline comparison of raw UWB vs UWB+Kalman; LiDAR zone-transition and stopping-distance tests with dynamic obstacles; and a sim-to-real comparison against the CW2 results. Mohammed Shalaby (M01035318) — ROS 2 architecture & systems integration. Research question: Which ROS 2 navigation and path-planning approach delivers reliable real-time following and return-to-home on constrained onboard hardware, and how does the integrated architecture perform under load? • Why this stack: justify the choice of ROS 2 Jazzy and the selected global and local planner/controller plugins against alternatives [cite: compare candidate planners/controllers], together with the rationale for the system architecture and state machine. • Validation: measure end-to-end latency, CPU/memory load on the Raspberry Pi, and node reliability/uptime; compare planner success-rate and path quality across candidate planners; and perform rigorous integration testing. Communicates directly with the supervisor on this strand. Vignesh Lakshmanasamy (M01026685) — Mechanical & power. Research question: How do the mechanical and power design choices affect load capacity, runtime, stability and energy efficiency, and which configurations are optimal for the target payload? • Why these choices: justify battery chemistry (LiFePO₄ vs Li-ion vs LiPo — energy density, safety, cycle life), motor and gear-ratio selection (torque vs speed for the rated load), chassis geometry (stability and weight) and wheel/caster choice (indoor traction) with comparisons and citations [cite: ...], not assumptions. • Validation: bench tests for load capacity, runtime under load, power-consumption profile and motor current/torque, plus stability tests; compare options experimentally where feasible and cite the literature for the rest.
Brief Methodology (Include Gantt chart)	The project follows a classic robotic system integration lifecycle: requirements refinement → mechanical/electrical fabrication → software porting and integration → iterative testing → evaluation and documentation.

	All development uses ROS 2 Jazzy on the target hardware stack (Raspberry Pi 5 as the onboard ROS 2 compute and an Arduino for low-level motor control); the Intel NUC and depth camera have been removed from scope. The team leverages the existing simulation codebase (github.com/cruisemaniac/Smart_Cart_With_ROS_PDE4435_CW2) for rapid iteration. Indicative schedule (detailed Gantt accompanies the build plan): • Wk 1-2: chassis fabrication, drivetrain and power assembly; ROS 2 skeleton and Arduino motor bring-up. • Wk 2-3: closed-loop drive + manual teleop on hardware; port three-zone LiDAR safety. • Wk 3-5: UWB-Kalman follow integration with the safety layer. • Wk 5-6: full integration and formal testing (sim-vs-real metrics). • Wk 6-7: demonstration capture and buffer; optional return-to-home. • Wk 7 → deadlines: report, blog and presentation preparation.
Evaluation Strategy	Following: distance-maintenance error ($1.0\text{ m} \pm 5\text{ cm}$), velocity smoothness and tracking accuracy (UWB + Kalman). Safety: collision/near-miss rate and successful zone transitions (warn/slow/stop) in dynamic obstacle tests. System: end-to-end latency, reliability/uptime, power consumption and simulation-to-real transfer performance. Metrics are collected via ROS 2 logging, RViz visualisation and scripted test scenarios in the MDX Robotics Lab. A short controlled trial with team members acting as the followed operator provides qualitative usability feedback. No external human participants are involved.
Ethical Declaration (i) or (ii) (Refer below)	(i) I have established that my study does not require additional human participation. I agree to re-apply for approval if the nature or goals of my project change. Reason: core work is technical prototype development and lab-based quantitative testing; only team members act as the followed operator. Any future trial with external participants would trigger re-application.
Deliverables (Bullet points)	• Fully functional physical Trolley-X prototype with integrated autonomous following and three-zone LiDAR safety on a modular payload platform. • Complete, reproducible ROS 2 Jazzy codebase (public GitHub). • Dissertation / technical report (IEEE format) including sim-to-real performance analysis. • 90-second demonstration video. • Revised detailed BOM, assembly instructions and evaluation data. • Open-source ROS 2 packages for the robotics community. • (If time permits) autonomous return-to-home path-planning demonstration.
Resources Needed	Hardware (self-funded total $\approx$ AED 6,440): aluminium chassis fabrication; modular payload platform; two high-torque DC gear motors with encoders; SmartDrive Duo MDDS30 motor driver; Slamtec RPLIDAR A1; UWB positioning kit (anchors + tag, e.g. ESP32-UWB / DW3000 $\times$3-4) for the physical follow-me function; Raspberry Pi 5 (16 GB) as onboard compute; Arduino for low-level motor control; LiFePO4 battery pack; DC-DC converters; power distribution and fusing; industrial E-stop; drive wheels; casters; wiring; and 3D-printed mounts. The Intel NUC 11 and Intel RealSense D415 have been removed from the BOM ($\approx$ AED 3,200 saving), reflecting the removal of vision-based item recognition from scope; a UWB positioning kit ($\approx$ AED 500), previously only simulated, has been added so the committed UWB follow-me function can run on real hardware.

	Software: ROS 2 Jazzy, Gazebo Harmonic, Python, Ubuntu and standard ROS 2 navigation/visualisation tooling. Facilities: Middlesex University Dubai Robotics & AI Lab. Funding: self-funded by the team; no external grant is required.
References	• Team's PDE4445 Coursework 2 simulation report and public GitHub repository. • ROS 2 and Nav2 documentation. • Literature on UWB + Kalman-filter sensor fusion for indoor localisation and follow-me robots. • Slamtec RPLIDAR A1 integration documentation. • Foxtech FOLO follow-cart product documentation (commercial comparator).

Signature: Ashwin / Shalaby / Vignesh

Date: 25-Jun-2026 (revised proposal)

Ethical Approval Guidelines

Declaration A

- (i) I have established that my study does not require additional human participation.
- (ii) I agree to re-apply for approval if the nature or goals of my project change.

Declaration B

Project goals involve human participation:

- (i) My study involves human participation through:
 - Observation
 - Questioning
- (ii) Participants will be selected without coercion.
- (iii) I will obtain written informed consent from each participant.
- (iv) I have arrangements in place for the protection of personal data.
- (v) I agree to re-apply for approval if the nature or goals of my project change.

To be completed by the supervisor:

Supervisor Name:

Is the proposal acceptable? (YES/NO):

If not, please provide feedback:

Signature:

Date:

Appendix: Explanation of Typical Terms Used in Research Proposals

- **Title:** Write a title which briefly describes the research problem and your approach to it.
- **Keywords:** To give a clear and concise description of the scope and nature of the report, such as the main variables to be considered.
- **Problem definition:** Correctly defining the problem is the crucial first step in the research process. If the research problem is defined incorrectly, the research objectives will also be wrong. Problems must be stated in terms of underlying causes - they must be structured in a way in which they can lead to a solution. It is critical that the statement be useful for development and evaluation of potential solutions.
- **Research Question:** Indicate what you want to know most and first out of your research. These are typically written in a 'question' format, for example, 'Does a robotic exoskeleton provide an efficient solution to physical rehabilitation of the upper arm for stroke patients?' Multiple research questions should be put in a list. Questions should not be answered by a simple YES or NO, they must generate a discussion.
- **Background Review:** Explain the technical/discipline area you will be working in, the problem area that you will be addressing in your research, and where you would locate your intended work in relation to previous researchers or existing solutions.
- **Aims:** They represent the changes you hope to achieve as a result of your work.
- **Objectives:** These are the activities you undertake and the methods you propose to bring these changes about. They should be in a bulleted list, and more specific than the aims. In fact, this list should be a way of breaking down the aims into more achievable tasks. Anywhere between 4 - 7 objectives are usually considered appropriate for a project of this scale.
- **Methodology:** A brief description of the steps to be followed to achieve the objectives that have been set out. This section should describe the development plan.
- **Gantt chart:** A Gantt chart is a horizontal bar chart used in project management to visually represent a project plan over time.
- **Evaluation Strategy:** Methods and techniques used in evaluating the project. There are two basic approaches, quantitative and qualitative.

Quantitative involves the generation of data, through experiments or simulation, in quantitative form, which can then be analysed.
Qualitative is concerned with subjective assessment of opinions, behaviour, impressions via, for example, interviews.

- **Deliverables:** Defining your intended outcomes (be as specific as possible). It is part of good project planning. Deliverables are linked to your aims.
- **Resource Needed:** Information on the hardware, software, and research resources you will be using during your research. Be specific
- Give the name of any software and hardware, rather than stating something generic.